

Jianye Xu

PhD Student.

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Research interests: Safe learning and distributed control, multi-agent decision making, and motion planning for connected and automated vehicles.

Education

RWTH Aachen University, Germany

2023-04 – present

- PhD, Chair of Embedded Software, Department of Computer Science
- *Thesis:* Learning Safe Coordination for Connected Vehicles (working title)
- *Advisor:* Prof. [Bassam Alrifaae](#)

RWTH Aachen University, Germany

2020-10 – 2022-09

- MSc, Automation Engineering, Institute of Automatic Control
- Overall average: 1.2/1.0 · *with distinction*
- *Thesis:* Parallel Priority-Based Trajectory Planning with Safety Guarantees for Networked Vehicles
- *Advisor:* Prof. [Stefan Kowalewski](#)

RWTH Aachen University, Germany

2019-10 – 2020-08

- BSc (Exchange Student), Mechanical Engineering, Faculty of Mechanical Engineering
- *Thesis:* Implementation and Comparison of Different Strategies for MTPA Control of Permanent-Magnet Synchronous Machines Considering Magnetic Saturation
- *Advisor:* Prof. [Jakob Andert](#)

Beijing Institute of Technology, China

2016-10 – 2019-09

- BSc, Vehicle Engineering, School of Mechanical Engineering
- Overall average: 91.48/100 · *Outstanding Graduate (top 5%)*
- *Advisor:* Prof. [Chen Lin](#)

Selected Publications

Journal Papers

- [J1] J. Beerwerth, **J. Xu**, S. Schäfer, F. Belderink, and B. Alrifaae. “Zero-Shot MARL Benchmark in the Cyber-Physical Mobility Lab”. In: *at - Automatisierungstechnik* 74.5 (2026), pp. 376–385. DOI: [10.1515/auto-2025-0057](#).
- [J2] **J. Xu**, J. Betz, A. Mokhtarian, A. Mittal, M. Cai, R. Mangharam, O. M. Shehata, C. M. Elias, J.-N. Zaeck, P. Scheffe, F. Jahncke, S. S. Ulhas, K. M. Arfvidsson, and B. Alrifaae. “Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms: Challenges and a Roadmap”. In: *IEEE Intelligent Transportation Systems Magazine* (2026), pp. 2–16. DOI: [10.1109/MITS.2026.3666681](#).
- [J3] A. Mokhtarian, **J. Xu**, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, V.-A. Le, S. Ulhas, J. Betz, S. Wilson, S. Berman, L. Paull, A. Prorok, and B. Alrifaae. “A Survey on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms: A Guide for Creating a New Testbed”. In: *IEEE Robotics & Automation Magazine* (2024), pp. 2–19. DOI: [10.1109/MRA.2024.3505772](#).

Conference Papers

- [C1] **J. Xu** and B. Alrifaae. “Beyond Safety Filtering: Control Barrier Function-Informed Reinforcement Learning for Connected and Automated Vehicles”. In: *2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)*, in Press. 2026.
- [C2] **J. Xu** and B. Alrifaae. “A Learning-Based Control Barrier Function for Car-Like Robots: Toward Less Conservative Collision Avoidance”. In: *2025 European Control Conference (ECC)*. 2025, pp. 988–995. DOI: [10.23919/ECC65951.2025.11187043](#).
- [C3] **J. Xu**, C. Che, and B. Alrifaae. “A Real-Time Control Barrier Function-Based Safety Filter for Motion Planning with Arbitrary Road Boundary Constraints”. In: *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*. 2025, pp. 2818–2825. DOI: [10.1109/ITSC60802.2025.11423203](#).
- [C4] P. Scheffe, **J. Xu**, and B. Alrifaae. “Limiting Computation Levels in Prioritized Trajectory Planning with Safety Guarantees”. In: *2024 European Control Conference (ECC)*. 2024, pp. 297–304. DOI: [10.23919/ECC64448.2024.10591179](#).
- [C5] **J. Xu**, P. Hu, and B. Alrifaae. “SigmaRL: A Sample-Efficient and Generalizable Multi-Agent Reinforcement Learning Framework for Motion Planning”. In: *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*. 2024, pp. 768–775. DOI: [10.1109/ITSC58415.2024.10919918](#).

Manuscripts Under Review

- [U1] **J. Xu** and B. Alrifaae. “TTCBF: A Truncated Taylor Control Barrier Function for High-Order Safety Constraints”. In: *arXiv preprint arXiv:2601.15196* (2026). DOI: [10.48550/arXiv.2601.15196](https://doi.org/10.48550/arXiv.2601.15196). arXiv: [2601.15196](https://arxiv.org/abs/2601.15196) [eess]. Under review at Automatica.

Awards

Best Poster Award	2025 German Autonomy Summit (GAS)	2025-12
Best Paper Nomination	IEEE International Conference on Intelligent Transportation Systems (ITSC)	2025-11
Dean’s List (top 5% in the academic year 2021/2022)	RWTH Aachen University	2023-01
MSc with distinction (GPA 1.2/1.0)	RWTH Aachen University	2022-09
German National Scholarship	RWTH Aachen University	2021-11
Patent: Quick Locking and Unlocking Device for Vehicle-Mounted Power Battery Box (first inventor, publication number: CN 111086379 B)	China National Intellectual Property Administration	2021-06
German National Scholarship	RWTH Aachen University	2020-11
Outstanding Graduate (GPA 91.48/100, top 5%)	Beijing Institute of Technology	2020-06
China National Scholarship	The Ministry of Education of the People’s Republic of China	2018-11
Second Prize in National Student Mechanical Product Digital Design Competition	China Mechanical Discipline Steering Committee	2018-06
Outstanding Undergraduate	Beijing Institute of Technology	2017-12
First Prize in Beijing Student Engineering Design Expression Competition	Beijing Municipal Commission of Education	2017-12
First Prize in National Student Drafting and Modelling Innovation Competition	China Cartographic Association	2017-07

Professional Activities

Program Committee

Associate Editor	IEEE International Conference on Intelligent Transportation Systems (ITSC)	2026
Invited Session Organizer	<i>IEEE ITSC, Naples, Italy</i> - Invited Session: “Bridging Optimal Control and Learning for Connected and Automated Vehicles” https://bassamlab.github.io/itsc2026-invited-session/	2026-09
Workshop Co-Organizer	<i>IEEE ITSC, Gold Coast, Australia</i> - Workshop: “Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI” https://www.itsc2025.mixedtraffic.org/	2025-11
Workshop Co-Organizer	<i>IEEE Intelligent Vehicles Symposium (IV) 2024, Jeju, South Korea</i> - Workshop: “1st Workshop on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms” https://cpm.lrt.unibw.de/workshop/	2024-06
Competition Co-Organizer	<i>RWTH Aachen University, Germany</i> - Competition: CPM Olympics (a motion planning competition for connected and automated vehicles) https://cpm.lrt.unibw.de/olympics/	2024-01 – 2024-06

Invited Talks

Workshop Invited Speaker	<i>IEEE ITSC, Naples, Italy</i> - Workshop: “The 16th Workshop & Industry Panel on Cooperative and Automated Driving and Future Mobility” https://ieee-itsc.org/2026/workshops/	2026-09
Workshop Invited Speaker	<i>IEEE ITSC, Gold Coast, Australia</i> - Workshop: “Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI” https://www.itsc2025.mixedtraffic.org/	2025-11

Journal Reviewing

Automatica, IEEE Control Systems Letters (L-CSS), IEEE Transactions on Intelligent Transportation Systems (T-ITS), IEEE Intelligent Transportation Systems Magazine (ITSM), IEEE Robotics and Automation Letters (RA-L)

Conference Reviewing

IEEE International Conference on Intelligent Transportation Systems (ITSC), IEEE Intelligent Vehicles Symposium (IV), European Control Conference (ECC), IEEE International Conference on Robotics and Automation (ICRA)

Membership

Graduate Student Member, IEEE, IEEE Intelligent Transportation Systems Society Membership, IEEE Robotics and Automation Society Membership, IEEE Control Systems Society Membership, IEEE Young Professionals

Teaching and Mentoring

Assistant Lecturer RWTH Aachen University, Germany

Winter terms 2024 & 2025

- Master Course: “Control and Perception in Networked and Autonomous Vehicles”
- <https://cpm.lrt.unibw.de/courses/master/>

Guest Lecturer RWTH Aachen University, Germany

2025-01

- Guest Lecture for [Master Course](#): “Control Barrier Functions and Their Applications to Collision Avoidance in Networked and Autonomous Vehicles”

Research Assistant RWTH Aachen University, Germany

2021-06 – 2022-02

- Institute for Plastics Processing (IKV)
- Developed a web-based interface for injection molding process optimization using JavaScript and Vue 3.
- Used Python for data processing and analysis, working with pandas and Keras.

Undergraduate Teaching Assistant Beijing Institute of Technology, China

2017-09 – 2017-10

- Delivered CAD software training to first-year students.

Supervised Theses RWTH Aachen University

2023 – 2026

- Ali Elbadawy (2026). *Integrating Barrier Functions into Multi-Agent Social Value Orientation for Unsignalized Intersections* (Master’s thesis).
- Massimo Marcon (2026). *Latent World Models for Motion Planning of Connected and Automated Vehicles* (Master’s thesis).
- Jiaxin Liu (2026). *Mixture of Expert Policies for Decision-Making and Motion Planning in Connected and Automated Vehicles* (Master’s thesis).
- George Halim (2026). *Barrier-Certified Exploration for Real-World Reinforcement Learning in the Cyber-Physical Mobility Lab* (Bachelor’s thesis).
- Yousef Sayed (2026). *A Study of Responsibility Assignment in Control Barrier Function for Connected and Automated Vehicles* (Bachelor’s thesis).
- Maximilian Petri (2025). *Reinforcement Learning for Formula Student Autonomous Racing* (Master’s thesis).
- Chang Che (2025). *Constraint-Enforced Multi-Agent Reinforcement Learning for Safe Motion Planning of Connected and Automated Vehicles* (Master’s thesis).
- Pan Hu (2025). *Situation-Aware Trajectory Planning of Connected and Automated Vehicles with Multi-Policy Reinforcement Learning* (Master’s thesis).
- Daniel Boules (2025). *A Prioritization-Based Approach for Enhanced Stability in Non-Stationary Multi-Agent Reinforcement Learning* (Bachelor’s thesis).
- Fynn Belderink (2025). *Model-Based Receding-Horizon Planning in Multi-Agent Reinforcement Learning* (Bachelor’s thesis).
- Yunjia Chen (2025). *Sensor Fusion-Enhanced Map Matching of Bicycle Routes Using Crowdsourced GPS and IMU Data from Smartphones* (Bachelor’s thesis).
- Xinyue Cui (2024). *Determining the Longitudinal and Lateral Acceleration of a Bicycle Using a Smartphone Mounted on Its Handlebar* (Master’s thesis).
- Lebo Liang (2024). *Safe Multi-Agent Control: Integrating Control Barrier Functions with Reinforcement Learning* (Bachelor’s thesis, Aachen University of Applied Sciences).
- Mohamed Shamekh (2024). *Model-Predictive-Control-Based Safe Multi-Agent Reinforcement Learning for Motion Planning* (Bachelor’s thesis).
- Omar Sobhy (2024). *Dealing with Non-Stationarity in Multi-Agent Reinforcement Learning Through Learning-Based Dynamic Prioritization* (Bachelor’s thesis).
- Xia Yan (2024). *Learning-Based Motion Planner for Connected and Automated Vehicles* (Master’s thesis).

- Xilei Chen (2024). *Imitation Learning from Optimization-Based Motion Planners for Connected and Automated Vehicles* (Bachelor's thesis).
- Raphael Malig (2024). *Investigating the Effect of Discretizing Continuous Action Space in Multi-Agent Reinforcement Learning for Motion Planning* (Bachelor's thesis).
- Andreas Pletschko (2024). *Leveraging Temporal Dependencies for Sample-Efficient and Generalizable Multi-Agent Reinforcement Learning in Motion Planning* (Bachelor's thesis).
- Ruizhang Zhou (2023). *AI-Based Generation of Testing Scenarios for Motion Planners on Connected and Automated Vehicles* (Master's thesis).